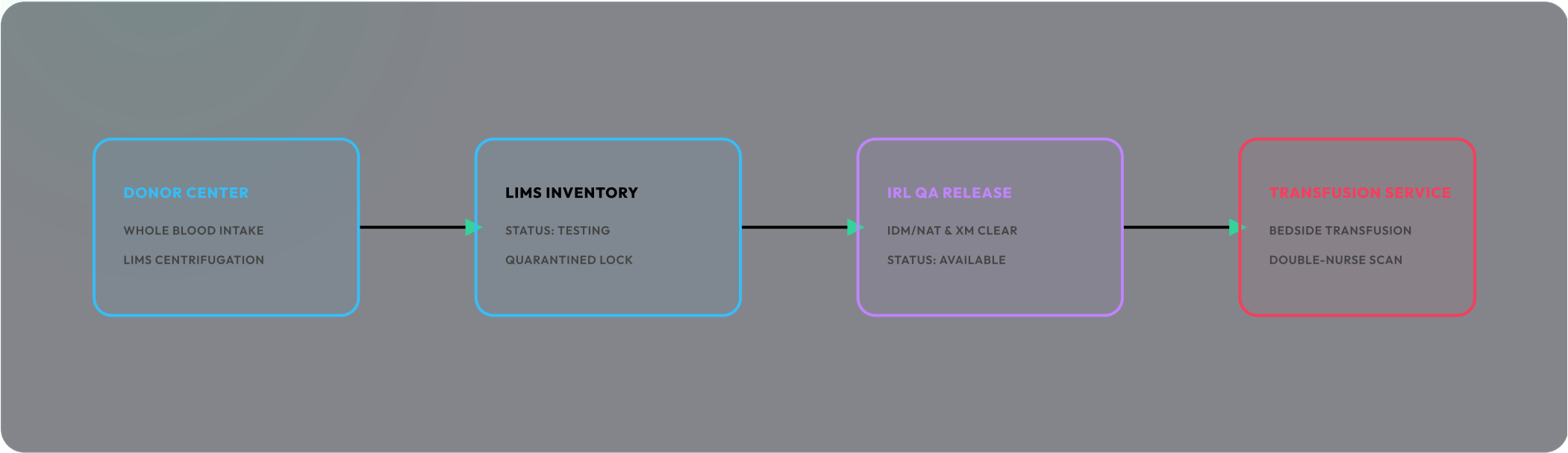


BLOOD BAG FLOWCHART

VN-BECS V2.0 SYSTEM INTEGRATION

DOCUMENT: BLOOD PRODUCT CHAIN
STATUS: **APPROVED**
REF: SOP 03 / SOP 08 / ISBT-128



STEP	SUBSYSTEM / FEATURE	WORKFLOW & INVENTORY TRACEABILITY LOGIC
01	Donor Center (Donor Center)	PHLEBOTOMY & COMPONENT SEPARATION Whole blood is collected from the donor. LIMS processes centrifugation to manufacture blood components (RBC, Platelets, Plasma), generating separate product codes under the same parent Donor Identification Number (DIN).

STEP	SUBSYSTEM / FEATURE	WORKFLOW & INVENTORY TRACEABILITY LOGIC
02	LIMS Quarantine Status	QUARANTINE STORAGE & TESTING STATUS Manufactured blood products are stored inside designated quarantine storage areas. The database strictly assigns the state 'TESTING' to prevent unauthorized allocation or release prior to final laboratory confirmation.
03	Immunohematology & Reference Lab (IRL)	QA RELEASE & AVAILABLE STATE TRANSITION The testing lab completes infectious disease marker checks (IDM/NAT) and ABO/Rh typing. Under strict QA authorization, clicking Quality Release automatically unlocks the corresponding blood units to 'AVAILABLE' status inside the Blood Supply Hub.
04	Blood Supply Hub (Hub) → Transfusion Service	FEFO DISPATCH, TRANSPORTATION, AND BEDSIDE SCAN The Blood Supply Hub selects blood bags using the FEFO algorithm, packaging them for IoT cold chain transportation to the hospital. Upon arrival, the Transfusion Service receives and verifies the units. Before transfusion, bedside nurses strictly scan the ISBT-128 DIN to ensure complete compatibility.